



# Perseverance reduces whereas impulsivity increases the risk of reoffending

## Verdicts as a source of information on psychological processing during real-life criminal action

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### Abstract

By law, every criminal verdict describes the norm violation in question. To date, the use of this information for prognostic purposes has been standardized only for sex offenses. This study presents a content-analytic approach to the material. A theory-based coding system allows for systematic tracing of the offender's affective and regulatory processing during norm violation, targeting the entire action sequence: initial activation affect (positive vs. negative), impulsivity vs. intentionality, affect regulation, and coping with intention-implementation difficulties. In a sample of  $N = 304$  males convicted of violent and/or sexual offenses, the processing variables were investigated with respect to (a) objective measurability, (b) relationships with established reoffending risk markers, and (c) their (incremental) performance in predicting reoffending. With respect to objectivity (a), the inter-rater agreement ( $n = 111$ ) varied widely among the variables (intraclass correlation range 0.03–0.62). Regarding risk markers (b), significant associations were found with the Level of Service Inventory–Revised (LSI-R) subscales *criminal history*, *alcohol/drugs*, *emotional/personal*, and *attitudes/orientation*. The Crime Scene Behavior Risk Measure for sexual offenses ( $n = 106$ ) correlated with psychological processing variables for the items *multiple juvenile offenders* and *male victim*. Two variables emerged as relevant for predicting recidivism (c): negatively motivated impulsivity predicted violent reoffending, while persistence when coping with implementation difficulties negatively predicted sexual and general reoffending. These relationships statistically mediated the predictive contribution of actuarial risk tools (LSI-R/crime scene behavior risk). Psychological processes during norm violation as reported in verdicts appear to represent underexploited information within risk assessment and beyond.

**Keywords** Criminal theory · Content analysis · Criminal verdicts · Crime Scene Behavior Risk Measure · Level of Service Inventory – Revised

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## Beharrlichkeit reduziert, Impulsivität erhöht das Risiko erneuter Straffälligkeit

Urteile als Informationsquelle für psychologische Prozesse während realer krimineller Handlungen

### Zusammenfassung

Aus rechtlichen Gründen enthält jedes Strafurteil eine Beschreibung der abgeurteilten Normverletzung. Bisher wird diese Information zu Prognosezwecken nur für Sexualstraftaten standardisiert genutzt. Die vorliegende Arbeit stellt einen inhaltsanalytischen Zugang zu Urteilstexten vor. Ein theoriebasiertes Kodiersystem ermöglicht es, die affektiven und regulativen Prozesse eines Täters während der Normverletzung systematisch nachzuzeichnen, wobei die gesamte Handlungssequenz erfasst wird: Aktivierungsaffekt (positiv vs. negativ), impulsive Reaktionsfreudigkeit vs. Absichtsbildung, Affektregulation und der Umgang mit Ausführungsproblemen. An einer Stichprobe wegen Gewalt- und/oder Sexualstraftaten verurteilter männlicher Jugendlicher und Erwachsener ( $N=304$ ) wurden die Prozessvariablen hinsichtlich (a) ihrer objektiven Messbarkeit, (b) ihrer Beziehungen zu etablierten Risikomarkern erneuter Straffälligkeit und (c) ihrer (inkrementellen) Leistung bei der Vorhersage erneuter Straffälligkeit untersucht. Die Objektivitätsanalyse (a) zeigte eine weite Fächerung der Inter-Rater-Übereinstimmungen ( $n=111$ ) zwischen den Variablen (Spannweite für den Intra-Klassen-Korrelationskoeffizient: 0,03–0,62). In Bezug auf die Risikomarker (b) zeigten sich signifikante Assoziationen mit den Teilskalen des Level of Service Inventory – Revised (LSI-R) *kriminelle Vorgeschichte*, *Alkohol/Drogen*, *emotional/persönlich* und *Einstellungen/Orientierung*. Der Tatbildrisikoscore (TBRS) für Sexualstraftaten ( $n=106$ ) korrelierte mit einigen psychologischen Verarbeitungsvariablen in den Items *mehrere jugendliche Täter* und *männliches Opfer*. Für die Rückfallvorhersage (c) erwiesen sich 2 Variablen als relevant: Negativ motivierte Impulsivität sagte gewalttätige Straffälligkeit vorher, während Beharrlichkeit bei Ausführungsproblemen sexuelle und allgemeine Straffälligkeit negativ vorhersagte. Diese Beziehungen mediierten statistisch den Vorhersagebeitrag aktuarischer Prognoseinstrumente (LSI-R/TBRS). Die psychologischen Prozesse während der Normverletzung, wie sie in Urteilen dargelegt werden, stellen sich als eine im Rahmen der Risikobewertung und darüber hinaus zu wenig genutzte Information dar.

**Schlüsselwörter** Kriminalitätstheorie · Inhaltsanalyse · Strafurteile · Tatbildrisikoscore · Level of Service Inventory—Revised

### Introduction

Originally invented for prognostic purposes, crime scene analysis has been successfully used as a prognostic tool for sex offenders (Biedermann 2013; Dahle et al. 2014; Lehmann et al. 2018). One obvious advantage of this approach is its immunity to certain biases, as it draws on behavioral traces documented in case files without relying on any kind of self-report. Crime scene behavior is assessed in terms of manifest behavior, typically using yes-or-no items. While the standardized assessment of crime scene behavior has improved the validity of risk assessment, objective data have the disadvantage of being rather distal to the actual criminogenic psychological processes. Even in a more explanatory variant of crime scene analysis (Biedermann 2013), the theoretical rationale remains rather vague as to which predictive, possibly criminogenic, differential psychological variables might be expressed in which type of observable crime scene behavior.

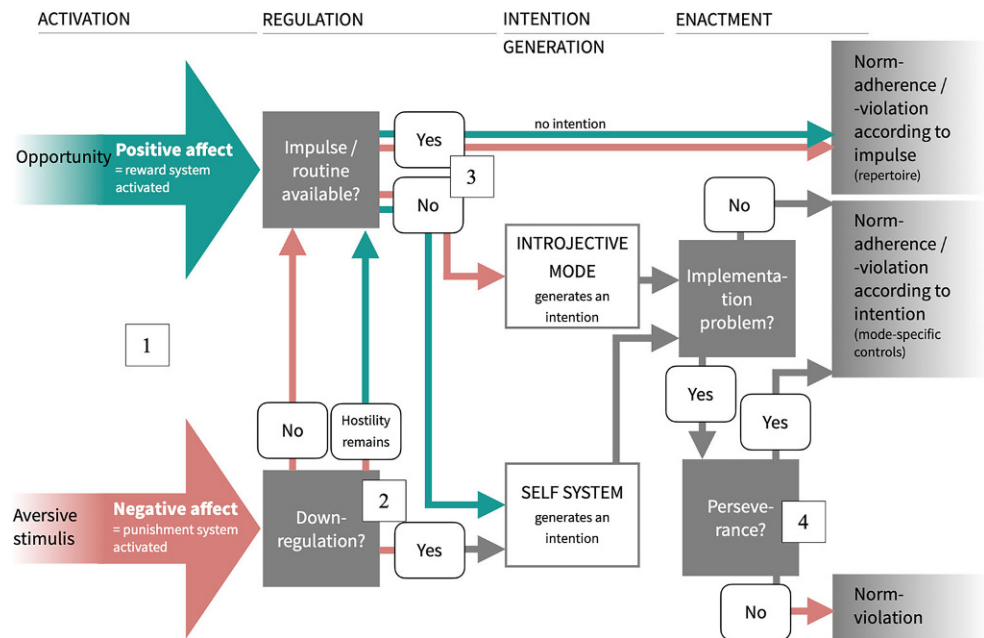
On a psychological level, every criminal action episode is a complex interaction between the to-be offender and his or her environment. To assess the psychological processes involved, taking into account sequentiality and chronology, a list of hard facts about the behavior at the crime scene is clearly insufficient. As certain legal systems such as that

of Germany require the consideration of these specific dynamics in the forensic assessment (Boetticher et al. 2019), the intuitive or “clinical” assessment has been transformed into the structured idiographic method, the core of which is the elaboration and justification of an individual theory of delinquency. This approach has been shown to be incrementally valid to a risk assessment based solely on actuarial measures (Dahle 2010). However, due to the importance of the individual theory of delinquency, the results are hardly comparable across individuals.

Nevertheless, nomothetic comparability and explanatory power are not necessarily mutually exclusive. In order to provide both, the analysis would have to be based on a general rather than an individual theory of the criminal action process. With this goal in mind, we have developed an action theory of criminal behavior based on basic psychological research and theory (primarily the personality systems interaction [PSI] theory of Julius Kuhl 2000, 2001; Kuhl et al. 2021). Our model will be outlined in its current state.

The self and affect regulation theory of adherence to norms (SARTAN; Fig. 1) is a theory of the actual genesis of adherence to social and legal norms. According to the theory, the psychological mode of both criminal and norm-conforming behavior is primarily determined by affect (e.g., emotions, moods). The affective state is constituted by the

**Fig. 1** The self and affect regulation theory of adherence to norms (SARTAN). Green arrows indicate positive affect (i.e., reward system activity), red arrows indicate negative affect (i.e., punishment system activity). Gray arrows indicate that no particular affective arousal is assumed or required. Note that hostility is treated as a special case of negative emotion, as it allows for a switch from negative to positive affect. The numbers indicate the psychological processes that were coded and will be referred to in the next section



level of positive affect (i.e., activation of the neural reward system, green in Fig. 1) and negative affect (i.e., activation of the neural punishment system, red in Fig. 1). The action sequence is thought to begin with *activation* followed by the phases of *regulation*, *intention generation*, and *enactment*. During these four phases, various external and internal processes are assumed to shape the final behavioral output by modifying the affective state. The specific mechanisms are rooted in PSI theory (Kuhl 2001; Kuhl et al. 2021). For instance, positive activation (first phase) as well as a successful downregulation (second phase) of negative activation affect are assumed to enable a mode of intention generation (third phase; see *self-mode* in Fig. 1) that is guided by the implicit *self-system*. According to Kuhl (2001), this *self-system* represents our personal experiences and relationships to the world in a network-like manner, including our personal morals, values, and preferences. However, negative affect is thought to deactivate the self-system (Kuhl 2001; Kuhl et al. 2021). Instead of emerging from this vast knowledge, under negative affect, an intention might form in a more analytic system (e.g., *intention memory*; see Baumann and Kuhl 2003; Kuhl et al. 2021). This mode of intention generation is called the *introjective mode* (Fig. 1) because it is assumed to make us susceptible to external cues and constraints, such as salient social norms and interpersonal expectations.<sup>1</sup> In addition, a third processing path through the SARTAN phases “skips” the intention-generation phase altogether; the result would be an impul-

sive response (see the red and green arrows at the top of Fig. 1). Finally, during enactment (fourth phase), obstacles may arise, depending on the situation and the quality of the action plan. These obstacles may be encountered either by sticking to the intention or by abandoning it, depending on the individual’s ability to endure the inhibition of positive affect (referred to as *frustration tolerance* by Kuhl, 2001).

Importantly, as Fig. 1 illustrates, the results of each stage of processing in SARTAN influence the next stage. For example, in the activation phase, positive or negative affect is induced. Only in the case of negative affect is an attempt at downregulation expected in the subsequent regulation phase. This logic results in a variety of possible processing paths.

SARTAN was formulated as an attempt to bridge the gap between (criminal) psychological theory and the prediction of individual criminal (re)offending. For prediction purposes, hypotheses about specific interactions can be derived from the theory. For example, personal commitment to legal norms (as part of the self-system; Fig. 1 and Kuhl 2001) should impact future legal behavior only if it is not rendered inaccessible by (increased susceptibility to) negative affect (Hamatschek et al. 2022).

The approach to criminal prognosis taken in this paper can be positioned between structured idiographic assessment and standardized crime scene-oriented behavioral analysis. The descriptions of the criminal acts in the respective verdicts are analyzed by means of a standardized, i.e., quantitative content analysis derived from SARTAN. The analysis system will be evaluated with respect to objectivity (or intersubjectivity) and various practically relevant markers of the statistical risk of reoffending. Finally, asso-

<sup>1</sup> The SARTAN distinction between *self-mode* and *introjective mode* corresponds to Kuhl’s (1996) distinction between *self-regulation* and *self-control*.

ciations with actual reoffending will be examined to assess the potential use of verdict-based content analysis for criminal prognostic purposes.

## Methods

### Sample

This study is based on a composite sample of incarcerated male offenders convicted of serious violent and/or sexual offenses. In total, data from  $N=304$  subjects were used, although some analyses were performed with subsamples.

### Sample generation

Most subjects were drawn from the CRIME II study (Dahle and Lehmann 2018), a retrospective longitudinal study conducted in the Berlin correctional system for male adults in the 2000s. From the 1995 to 1998 release cohorts of all Berlin prisons, all inmates convicted of a sexual or homicide offense were included in the original study. In addition, a random sample of inmates serving a sentence of at least 3 years for a violent offense was added. Of the 221 subjects in the original CRIME II sample,  $n=218$  subjects were included in this analysis, while three subjects' verdicts were illegible or unavailable. A smaller subsample was obtained from the CRIME III study (Dahle and Schmidt 2014), which used the same methodological approach as its predecessor, CRIME II, but was conducted with young inmates released from Berlin juvenile prisons between 1998 and 2002. Unfortunately, sentences were only available for a minority of the original sample, so only  $n=86$  of the 261 cases could be included.

### Sample description

The age of the subjects at the time of the index offenses ranged from 14:02 to 69:00 years (mean [ $M$ ]=27:08; standard deviation [ $SD$ ]=9:09). Most subjects (176; 57.9%) had been convicted of a nonsexual violent offense (including murder), 128 subjects (42.1%) had committed a sexual offense. In cases where multiple offenses were adjudicated in the index conviction, they were categorized into these offense types based on the most serious offense (as judged by the sentencing range).

### Variables

Several variables related to affective processing and self-regulation during the index offense were measured using the content-analytic coding system URD (*Urteilstextbasierte Rekonstruktion des Deliktablaufs*, which is German for ver-

*dict-based reconstruction of the offense sequence*). Additionally, subjects' ratings on the *Level of Service Inventory-Revised* (LSI-R; Dahle et al. 2012) and, for sex offenders, the *Crime Scene Behavior Risk Measure* (CBR; Dahle et al. 2014) were used.

### Psychological processes during criminal behavior

Based on the descriptions of the sequence of events and the offender's internal state as formulated in the verdicts, the URD guides the tracing of the processing path through the SARTAN action phases (Fig. 1). It consists of a nine-page step-by-step coding system with detailed operationalizations and examples.<sup>2</sup>

### Coding procedure

In German criminal judgments, a description of the specific criminal act of interest can be found in the "facts of the case" section (*Sachverhalt*; typically, 4. II). This section was the sole source of information for all URD codings. Many verdicts described a single offense, while others described a series of offenses. All offense descriptions in a verdict were coded and then analyzed as one case (i.e., each judgement constituted one case). In total, the URD codings were performed by 14 different coders. The coders were psychology students working on their B.Sc. (three coders) or M.Sc. (10 coders) theses, as well as the first author who also trained each of the other coders in at least three sessions using case material.

### URD variables

The URD provides one or more strings of code per offense, with each string representing an action unit. Only the action units that contain norm-violating elements<sup>3</sup> are then counted, aggregated, and converted for further analysis. In this way, only the processing during the norm-violating action is included.

The variables of interest for this study relate to four processes that are central according to SARTAN and are numbered 1 to 4 in Fig. 1: activation (via positive or negative affect), regulation of negative affect (downregulation vs. no downregulation vs. only hostile emotions persisting), impulsive reactivity, and perseverance in the face of problems and obstacles during behavioral enactment. Since the first three processes can occur only once per activation, their frequency was divided by the number of total activations. The

<sup>2</sup> The codebook and additional information can be obtained from the first author.

<sup>3</sup> Violations of legal norms in intentions, impulses, and behavioral outcomes were all counted as norm-violating elements.

resulting variables were named *activation affect A*, *down-regulation A*, *hostility A*, and *impulse A*, where “A” is short for “per activation.”

However, due to the path dependencies in the coding procedure, an additional alternative transformation seemed reasonable. For all variables in the second phase (*regulation*) or later, the frequencies were divided by their respective preconditions in order to isolate the specific psychological processes. For example, the number of downregulations of negative affect was divided by the number of instances of negative affect to obtain a measure of net downregulation of negative affect, independent of its rate of occurrence: *downregulation NA* (“NA” indicating “per negative activation”). Similarly, the variables *hostility NA*, *impulse NA*, *impulse PA*, and *impulse H* were created (“PA” indicating “per positive activation”; “H” indicating “per hostility coding”). Only one variable was coded for the enactment phase (number 4 in Fig. 1): the perseverance in the face of implementation difficulties or *perseverance EP* (“EP” indicating “per enactment problem”). Importantly, all precondition-related variables (i.e., those ending in NA, PA, H, or EP) are defined only for cases with at least one instance of the respective precondition. This results in different sample sizes in the different analyses.

An example of the coding and aggregation process can be found in the appendix.

### Personal risk factors

The LSI-R (Dahle et al. 2012; see Andrews and Bonta 1995 for the Canadian original) is a 54-item instrument that assesses the risk of reoffending. It contains both static and dynamic risk factors. The LSI-R has been shown to predict not only general reoffending (Fazel et al. 2022), but also specifically violent reoffending (e.g., Barnoski and Aos 2003; Dahle 2006). In contrast, it does not appear to be a useful predictor of sexual reoffending (Ragusa-Salerno et al. 2013). The LSI-R was scored by psychologists for all subsamples as part of the respective original studies. The total score and its 10 subscores were used for this analysis. LSI-R scores were available for 274 of the 304 subjects.

### Crime scene behavior

The CBR (Dahle et al. 2014) is a standardized instrument containing seven items about specific aspects of a sexual offense. The CBR sum score is used as a tool to predict sexual reoffenses. CBR data were only available for 106 of the 304 subjects, primarily because the instrument is only applicable to sexual offenses.

## Reoffending

Reoffending was measured on the basis of official criminal records, generating three dichotomous variables: *general reoffense* was operationalized by a new prison sentence for any type of offense (realized in 36.5% of cases), *violent reoffense* by any reconviction for a violent offense (34.3%), and *sexual reoffense* by any reconviction for a sexual offense (9.7%), all within 5 years of release. Reoffending data were missing for 27 CRIME II subjects, so the sample for these analyses was reduced accordingly.

## Analysis steps

### 1. Objectivity

Inter-rater reliability was calculated using the subsample of  $n=111$  subjects for whom there were different URD codings of the same judgment (by two,  $n=89$ ; three,  $n=19$ ; or four,  $n=3$  different coders).

All subsequent analyses were performed on a dataset containing only one coding per case. For the 111 cases coded by multiple coders, one coding was randomly selected and the others were excluded.

### 2. Associations with risk markers

The correlations of the URD variables with the LSI-R total score and subscores were analyzed to get an idea of which personal characteristics might be related to psychological processing during criminal behavior. In addition to these exploratory analyses, two hypotheses were derived from the SARTAN assumption that unregulated negative affect impedes the self-system mode of intention generation (see Fig. 1). LSI-R subscale 10 (*attitudes/orientation*) was assumed to be at least partially indicative of implicit beliefs stored in the self-system (see Kuhl 2001) that are favorable to crime. Since, according to SARTAN, these beliefs would only exert their criminogenic influence in the absence of negative affect, and since only cases of criminal (as opposed to noncriminal) behavior were part of the analysis, it was expected that the LSI-R subscale 10 would correlate positively with both *positive activation A* (H1) and *down-regulation NA* (H2).

For the sex offenders in the sample, the correlations of the URD variables with the CBR total score and its seven items were analyzed to infer which objective characteristics of the criminal act might be related to psychological processes. *Impulse H* was excluded from this analysis because it was computable only for 13 sex offenders, as only their offenses were coded with at least one instance of hostility.

### 3. Predicting reoffending

The utility of the URD variables as predictors of reoffending was assessed using logistic regression. First, for each URD variable, we investigated whether it predicted the reoffense criteria. If so, the incremental predictive contribution was assessed by fitting the following models:

- *For the violent reoffense criterion:* The *index offense type* (0 = violent, 1 = sexual) was introduced as a second predictor to examine whether the respective URD predictors tap into offense-specific psychological processing tendencies that provide a predictive advantage over the straightforward approach of predicting future behavior from past behavior. If the URD predictors retained statistical significance, the LSI-R sum score was added as a third predictor.
- *For the sexual reoffense criterion:* The second model included the same predictors as for *violent reoffense*. Then, if the URD predictor retained significance, the CBR was added to the model (instead of the LSI-R, which does not predict sexual offending, at least for sex offenders; see Ragusa-Salerno et al. 2013). As the CBR only applies to sexual offenses, only offenders initially convicted of a sexual offense were included in this third model. The now obsolete predictor *index offense type* was omitted.
- *For the general reoffense criterion:* Only one additional model was fitted, including the LSI-R total score as a co-predictor.

## Results

### 1. Objectivity

Inter-rater reliability was estimated from the multiple-coded cases using intraclass correlations. The results are displayed in Table 1. They vary widely. While the downregulation variables, *impulse A*, *impulse PA*, *impulse H*, and *hostility NA*, showed poor inter-rater agreement, the other variables reached a fair agreement of intraclass correlations ( $ICC \geq 0.40$ ) according to the conventions proposed by Cicchetti (1994). For *perseverance EP*, the agreement was good ( $ICC \geq 0.60$ ). With reference to the stricter classification proposed by Koo and Li (2016), only *positive activation A* and *perseverance EP* were coded with at least moderate agreement ( $ICC \geq 0.50$ ); and the ICC was significantly  $\geq 0.50$  only for the latter).

### 2. Associations with risk markers

For correlation tests involving the URD variables, Kendall's tau was used as a nonparametric alternative to Pearson's  $r$ .

The correlations of the URD variables with the LSI-R sum score and subscales are presented in Table 2. The LSI-R sum score was uncorrelated with the URD variables except for *downregulation NA*,  $\tau = -0.14$ ,  $p = 0.02$ . The strongest associations were found between *downregulation NA* and the *alcohol/drugs* subscale,  $\tau = -0.24$ ,  $p < 0.001$ , and between *downregulation NA* and the *emotional/personal*

**Table 1** ICCs for the URD variable codings

|                                   | Positive activation<br>A | Downregulation |      | Hostility |      | Impulse |      |      |      | Perseverance |  |
|-----------------------------------|--------------------------|----------------|------|-----------|------|---------|------|------|------|--------------|--|
|                                   |                          | A              | NA   | A         | NA   | A       | PA   | NA   | H    | EP           |  |
| ICC (1,1)                         | 0.53                     | 0.15           | 0.22 | 0.44      | 0.33 | 0.29    | 0.03 | 0.44 | 0.33 | 0.62         |  |
| bound <sub>95%</sub> <sup>a</sup> | 0.49                     | 0.13           | 0.19 | 0.41      | 0.30 | 0.26    | 0.02 | 0.41 | 0.29 | 0.59         |  |

<sup>a</sup>Boundary of upper one-sided 95% confidence interval

**Table 2** Kendall's  $\tau$  correlations between psychological processes during criminal behavior and LSI-R

| URD variable                 | <i>n</i>  | 1      | 2      | 3     | 4     | 5     | 6     | 7     | 8     | 9      | 10     | LSI-R  |
|------------------------------|-----------|--------|--------|-------|-------|-------|-------|-------|-------|--------|--------|--------|
| <i>Positive activation A</i> | 304       | 0.07   | 0.00   | 0.02  | -0.04 | -0.08 | -0.01 | -0.02 | -0.04 | 0.00   | 0.18*  | 0.02   |
| <i>Downregulation</i>        | <i>A</i>  | 304    | -0.10* | -0.00 | 0.02  | -0.02 | -0.01 | 0.07  | 0.06  | -0.12* | -0.11* | -0.10  |
|                              | <i>NA</i> | 172    | -0.07  | -0.05 | -0.01 | -0.10 | -0.10 | 0.03  | -0.01 | -0.24* | -0.18* | 0.03   |
| <i>Hostility</i>             | <i>A</i>  | 304    | -0.05  | 0.01  | 0.03  | 0.04  | 0.04  | 0.00  | -0.03 | 0.10*  | 0.03   | -0.14* |
|                              | <i>NA</i> | 172    | -0.00  | 0.04  | 0.06  | 0.01  | 0.01  | 0.03  | -0.03 | 0.12   | 0.06   | -0.05  |
| <i>Impulse</i>               | <i>A</i>  | 304    | 0.03   | 0.00  | 0.01  | 0.09  | 0.04  | -0.04 | 0.07  | 0.11*  | 0.01   | -0.08  |
|                              | <i>PA</i> | 189    | 0.11   | -0.02 | -0.11 | 0.06  | -0.05 | 0.05  | 0.11  | 0.10   | 0.02   | -0.07  |
|                              | <i>NA</i> | 172    | 0.03   | -0.00 | 0.00  | 0.10  | -0.00 | -0.04 | 0.02  | 0.04   | 0.06   | -0.03  |
|                              | <i>H</i>  | 68     | -0.01  | 0.04  | 0.06  | -0.08 | 0.05  | -0.10 | 0.11  | -0.05  | -0.15  | 0.08   |
| <i>Perseverance EP</i>       | 213       | -0.12* | 0.02   | -0.03 | 0.06  | 0.09  | -0.02 | -0.05 | -0.03 | 0.03   | 0.03   | -0.04  |

LSI-R (*Level of Service Inventory – Revised*) subscales: 1 criminal history, 2 education/employment, 3 financial, 4 family/marital, 5 accommodation, 6 leisure/recreation, 7 companions, 8 alcohol/drugs, 9 emotional/personal, 10 attitudes/orientation

\*Indicates statistical significance at  $p < 0.05$

**Table 3** Kendall's  $\tau$  correlations between psychological processes during sexual offenses and Crime Scene Behavior Risk Measure (CBR)

| URD variable                 | <i>n</i> | 1. Multiple juvenile offenders | 2. Explicit planning | 3. Actively seeking victim | 4. Approach-explicit | 5. Male victim | 6. Hands off: victim active | 7. Sexualized language | CBR   |
|------------------------------|----------|--------------------------------|----------------------|----------------------------|----------------------|----------------|-----------------------------|------------------------|-------|
| <i>Positive activation A</i> | 106      | 0.17                           | 0.00                 | 0.01                       | 0.05                 | 0.11           | 0.02                        | 0.04                   | 0.11  |
| <i>Downregulation</i>        |          |                                |                      |                            |                      |                |                             |                        |       |
| <i>A</i>                     | 106      | -0.31*                         | -0.03                | 0.04                       | 0.08                 | -0.08          | 0.05                        | 0.06                   | -0.02 |
| <i>NA</i>                    | 33       | -0.24                          | -0.00                | 0.10                       | 0.16                 | 0.02           | 0.01                        | 0.03                   | 0.07  |
| <i>Hostility</i>             |          |                                |                      |                            |                      |                |                             |                        |       |
| <i>A</i>                     | 106      | -0.10                          | -0.09                | -0.07                      | -0.12                | -0.02          | -0.07                       | -0.03                  | -0.16 |
| <i>NA</i>                    | 33       | 0.09                           | -0.01                | -0.08                      | -0.15                | 0.13           | 0.07                        | 0.16                   | 0.04  |
| <i>Impulse</i>               |          |                                |                      |                            |                      |                |                             |                        |       |
| <i>A</i>                     | 106      | -0.20*                         | -0.15                | -0.01                      | -0.08                | 0.09           | -0.08                       | 0.08                   | -0.08 |
| <i>PA</i>                    | 94       | -0.15                          | -0.16                | -0.04                      | -0.03                | 0.10           | -0.06                       | 0.03                   | -0.09 |
| <i>NA</i>                    | 33       | 0.06                           | 0.18                 | -0.09                      | -0.11                | 0.47*          | -0.04                       | -0.09                  | 0.26  |
| <i>Perseverance EP</i>       | 87       | -0.01                          | -0.04                | -0.12                      | -0.14                | -0.24*         | 0.06                        | 0.09                   | -0.14 |

\*Indicates statistical significance at  $p < 0.05$ 

subscale,  $\tau = -0.18$ ,  $p = 0.01$ . There was also the expected association between *positive activation A* and the *attitudes/orientation* subscale,  $\tau = 0.18$ ,  $p < 0.001$  (one-sided), confirming H1. There was no evidence of the expected association between *downregulation NA* and the *attitudes/orientation* subscale (H2),  $\tau = 0.03$ ,  $p = 0.34$  (one-sided). Overall, the URD variables showed quite distinct patterns of association with the different risk factors, although *hostility NA*, *impulse PA*, *impulse NA*, and *impulse H* showed no association at all.

The correlations of the URD variables with the CBR score and its individual items are displayed in Table 3. The total CBR score did not show any associations with the URD variables. Associations were found between *impulse NA* and *male victim*,  $\tau = 0.47$ ,  $p = 0.01$ , between *downregulation A* and *multiple offenders*,  $\tau = -0.31$ ,  $p = 0.001$ , between *perseverance EP* and *male victim*,  $\tau = -0.24$ ,  $p = 0.01$ , and between *impulse A* and *multiple juvenile offenders*,  $\tau = -0.20$ ,  $p = 0.04$ .

### 3. Predicting reoffending

First, it was tested whether the reference risk assessment instruments showed the necessary associations with reoffending. The LSI-R correlated with general reoffense,  $r = 0.24$ ,  $p < 0.001$ , and violent reoffense,  $r = 0.19$ ,  $p = 0.002$ , and for the sex offenders ( $n = 106$ ), the CBR correlated with sexual reoffense,  $r = 0.24$ ,  $p = 0.01$ .

The indices of the logistic regression models predicting reoffending are displayed in Table 4.

#### Predicting general reoffending

One URD variable emerged as a significant predictor: *Perseverance EP* statistically decreased the risk of reoffending,  $\beta = -0.94$ ,  $z = -2.50$ ,  $p = 0.01$ , and it still did so when controlling for LSI-R score,  $\beta = -1.00$ ,  $z = -2.53$ ,  $p = 0.01$ .

#### Predicting violent reoffending

Violent reoffending was positively predicted by all types of impulsivity except *impulse PA*:  $\beta = 0.88$ ,  $z = 2.13$ ,  $p = 0.03$  for *impulse A*,  $\beta = 1.34$ ,  $z = 2.35$ ,  $p = 0.02$  for *impulse NA*, and  $\beta = 1.35$ ,  $z = 2.01$ ,  $p = 0.04$  for *impulse H*. These predictive relationships persisted when controlling for index offense type:  $\beta = 0.96$ ,  $z = 2.29$ ,  $p = 0.02$  for *impulse A*,  $\beta = 1.47$ ,  $z = 2.51$ ,  $p = 0.01$  for *impulse NA*, and  $\beta = 1.74$ ,  $z = 2.34$ ,  $p = 0.02$  for *impulse H*. Finally, the impulsivity variables showed incremental prediction with respect to the LSI-R:  $\beta = 0.96$ ,  $z = 2.23$ ,  $p = 0.03$  for *impulse A*,  $\beta = 1.42$ ,  $z = 2.40$ ,  $p = 0.02$  for *impulse NA*, and  $\beta = 2.53$ ,  $z = 2.55$ ,  $p = 0.01$  for *impulse H*. Interestingly, in the latter two models, the LSI-R was no longer a significant predictor

**Table 4** Indices for logistic regression models predicting reoffenses

| Psychological processes                           | Criterion                |      |        |                            |      |        |                           |      |        |
|---------------------------------------------------|--------------------------|------|--------|----------------------------|------|--------|---------------------------|------|--------|
|                                                   | <i>General reoffense</i> |      |        | <i>Violent (re)offense</i> |      |        | <i>Sexual (re)offense</i> |      |        |
|                                                   | $\beta$                  | SE   | z      | $\beta$                    | SE   | z      | $\beta$                   | SE   | z      |
| <b>Activation, N = 276</b>                        |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.68                    | 0.19 | -3.55* | -0.74                      | 0.19 | -3.83* | -3.03                     | 0.42 | -7.24* |
| Positive activation A                             | 0.23                     | 0.27 | 0.86   | 0.17                       | 0.28 | 0.62   | 1.28                      | 0.50 | 2.55*  |
| Intercept                                         | —                        | —    | —      | —                          | —    | —      | -3.40                     | 0.47 | -7.20* |
| Positive activation A                             | —                        | —    | —      | —                          | —    | —      | 0.34                      | 0.59 | 0.58   |
| Offense type                                      | —                        | —    | —      | —                          | —    | —      | 1.65                      | 0.55 | 2.97*  |
| <b>Downregulation, N = 276</b>                    |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.48                    | 0.14 | -3.38* | -0.55                      | 0.14 | -3.83* | -1.96                     | 0.21 | -9.23* |
| Downregulation A                                  | -0.39                    | 0.35 | -1.09  | -0.53                      | 0.37 | -1.45  | -2.51                     | 1.21 | -2.08* |
| Intercept                                         | —                        | —    | —      | —                          | —    | —      | -2.95                     | 0.43 | -6.88* |
| Downregulation A                                  | —                        | —    | —      | —                          | —    | —      | -1.99                     | 1.25 | -1.60  |
| Offense type                                      | —                        | —    | —      | —                          | —    | —      | 1.60                      | 0.49 | 3.30*  |
| <b>Downregulation if negative affect, N = 173</b> |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.69                    | 0.23 | -3.06* | -0.47                      | 0.22 | -2.17* | -2.43                     | 0.39 | -6.21* |
| Downregulation NA                                 | -0.07                    | 0.36 | -0.21  | -0.61                      | 0.37 | -1.66  | -0.49                     | 0.70 | -0.71  |
| <b>Hostility, N = 276</b>                         |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.52                    | 0.14 | -3.75* | -0.65                      | 0.14 | -4.62  | -2.10                     | 0.21 | -9.78* |
| Hostility A                                       | -0.27                    | 0.40 | -0.68  | -0.03                      | 0.39 | -0.08  | -1.17                     | 0.89 | -1.31  |
| <b>Hostility if negative affect, N = 173</b>      |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.72                    | 0.21 | -3.46* | -0.78                      | 0.21 | -3.71* | -2.73                     | 0.41 | -6.72* |
| Hostility NA                                      | -0.00                    | 0.38 | -0.01  | 0.18                       | 0.38 | 0.47   | 0.33                      | 0.69 | 0.48   |
| <b>Impulsivity, N = 276</b>                       |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.58                    | 0.14 | -4.20* | -0.78                      | 0.14 | -5.47* | -2.13                     | 0.22 | -9.78* |
| Impulse A                                         | 0.19                     | 0.41 | 0.46   | 0.88                       | 0.41 | 2.13*  | -0.84                     | 0.86 | -0.98  |
| Intercept                                         | —                        | —    | —      | -0.91                      | 0.19 | -4.81* | —                         | —    | —      |
| Impulse A                                         | —                        | —    | —      | 0.96                       | 0.42 | 2.29*  | —                         | —    | —      |
| Offense type                                      | —                        | —    | —      | 0.28                       | 0.26 | 1.07   | —                         | —    | —      |
| <b>N = 272</b>                                    |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | —                        | —    | —      | -2.13                      | 0.46 | -4.66* | —                         | —    | —      |
| Impulse A                                         | —                        | —    | —      | 0.96                       | 0.43 | 2.23*  | —                         | —    | —      |
| Offense type                                      | —                        | —    | —      | 0.35                       | 0.27 | 1.29   | —                         | —    | —      |
| Level of Service Inventory-Revised                | —                        | —    | —      | 0.05                       | 0.02 | 2.98*  | —                         | —    | —      |
| <b>Impulsivity if positive affect, N = 168</b>    |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.49                    | 0.17 | -2.93* | -0.55                      | 0.17 | -3.25* | -1.84                     | 0.24 | -7.80* |
| Impulse PA                                        | -0.26                    | 0.65 | -0.41  | -0.20                      | 0.65 | -0.31  | -0.74                     | 1.15 | -0.65  |
| <b>Impulsivity if negative affect, N = 159</b>    |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | -0.82                    | 0.19 | -4.40* | -0.90                      | 0.19 | -4.75* | -2.75                     | 0.36 | -7.64* |
| Impulse NA                                        | 0.97                     | 0.56 | 1.73   | 1.34                       | 0.57 | 2.35*  | 0.37                      | 1.03 | 0.36   |
| Intercept                                         | —                        | —    | —      | -1.01                      | 0.22 | -4.62* | —                         | —    | —      |
| Impulse NA                                        | —                        | —    | —      | 1.47                       | 0.59 | 2.51*  | —                         | —    | —      |
| Offense type                                      | —                        | —    | —      | 0.48                       | 0.43 | 1.13   | —                         | —    | —      |
| <b>N = 157</b>                                    |                          |      |        |                            |      |        |                           |      |        |
| Intercept                                         | —                        | —    | —      | -1.90                      | 0.55 | -3.45* | —                         | —    | —      |
| Impulse NA                                        | —                        | —    | —      | 1.42                       | 0.59 | 2.40*  | —                         | —    | —      |
| Offense type                                      | —                        | —    | —      | 0.50                       | 0.43 | 1.16   | —                         | —    | —      |
| Level of Service Inventory-Revised                | —                        | —    | —      | 0.04                       | 0.02 | 1.86   | —                         | —    | —      |

**Table 4** (Continued)

| Psychological processes                        | Criterion                |           |          |                            |           |          |                           |           |          |
|------------------------------------------------|--------------------------|-----------|----------|----------------------------|-----------|----------|---------------------------|-----------|----------|
|                                                | <i>General reoffense</i> |           |          | <i>Violent (re)offense</i> |           |          | <i>Sexual (re)offense</i> |           |          |
|                                                | $\beta$                  | <i>SE</i> | <i>z</i> | $\beta$                    | <i>SE</i> | <i>z</i> | $\beta$                   | <i>SE</i> | <i>z</i> |
| <b>Impulsivity if hostility, <i>N</i> = 58</b> |                          |           |          |                            |           |          |                           |           |          |
| Intercept                                      | -0.97                    | 0.39      | -2.46*   | -1.03                      | 0.40      | -2.59*   | -2.14                     | 0.58      | -3.66*   |
| Impulse H                                      | 0.64                     | 0.67      | 0.95     | 1.35                       | 0.68      | 2.01*    | -0.68                     | 1.24      | -0.55    |
| Intercept                                      | —                        | —         | —        | -1.56                      | 0.51      | -3.07*   | —                         | —         | —        |
| Impulse H                                      | —                        | —         | —        | 1.74                       | 0.74      | 2.34*    | —                         | —         | —        |
| Offense type                                   | —                        | —         | —        | 1.47                       | 0.70      | 2.09*    | —                         | —         | —        |
| Intercept                                      | —                        | —         | —        | -3.41                      | 1.21      | -2.81*   | —                         | —         | —        |
| Impulse H                                      | —                        | —         | —        | 1.91                       | 0.78      | 2.45*    | —                         | —         | —        |
| Offense type                                   | —                        | —         | —        | 1.68                       | 0.73      | 2.29*    | —                         | —         | —        |
| Level of Service Inventory-Revised             | —                        | —         | —        | 0.07                       | 0.04      | 1.79     | —                         | —         | —        |
| <b>Perseverance, <i>N</i> = 188</b>            |                          |           |          |                            |           |          |                           |           |          |
| Intercept                                      | 0.04                     | 0.26      | 0.17     | -0.82                      | 0.28      | -2.95*   | -1.25                     | 0.33      | -3.75*   |
| Perseverance EP                                | -0.94                    | 0.38      | -2.50*   | 0.35                       | 0.38      | 0.91     | -2.21                     | 0.69      | -3.20*   |
| Intercept                                      | —                        | —         | —        | —                          | —         | —        | -3.60                     | 1.03      | -3.51*   |
| Perseverance EP                                | —                        | —         | —        | —                          | —         | —        | -2.33                     | 0.74      | -3.17*   |
| Offense type                                   | —                        | —         | —        | —                          | —         | —        | 3.24                      | 1.05      | 3.08*    |
| <b><i>N</i> = 186</b>                          |                          |           |          |                            |           |          |                           |           |          |
| Intercept                                      | -1.37                    | 0.57      | -2.38*   | —                          | —         | —        | -0.157                    | 1.00      | -1.57    |
| Perseverance EP                                | -1.00                    | 0.40      | -2.53*   | —                          | —         | —        | -2.52                     | 0.83      | -3.02*   |
| Level of Service Inventory-Revised             | 0.06                     | 0.02      | 2.88*    | —                          | —         | —        | —                         | —         | —        |
| Crime Scene Behavior Risk Measure              | —                        | —         | —        | —                          | —         | —        | 0.43                      | 0.29      | 1.50     |
| <b><i>N</i> = 81 (only sex offenders)</b>      |                          |           |          |                            |           |          |                           |           |          |

Differences in sample sizes are due to missing reoffense data and some psychological processing variables not being assessable in every case

Offense type coding: 0 violent, 1 sexual

\*Indicates statistical significance at  $p < 0.05$

of violent reoffending. Its predictive contribution appeared to be completely mediated by *impulse NA* or *impulse H*, respectively.<sup>4</sup>

### Predicting sexual reoffending

Sexual reoffending was positively predicted by *positive activation A*,  $\beta = 1.28$ ,  $z = 2.55$ ,  $p = 0.01$ , and negatively predicted by *downregulation A*,  $\beta = -2.51$ ,  $z = -2.08$ ,  $p = 0.04$ , and *perseverance EP*,  $\beta = -2.21$ ,  $z = -3.20$ ,  $p = 0.001$ . However, only *perseverance EP* retained significance when controlling for index offense type,  $\beta = -2.33$ ,  $z = -3.17$ ,  $p = 0.002$ . Within the sex offender subsample, it also showed an incremental predictive contribution with respect to CBR,  $\beta = -2.52$ ,  $z = -3.02$ ,  $p = 0.002$ . The CBR score was no longer a significant predictor in this model.

<sup>4</sup> To ensure that the mediation was truly due to *impulse NA/H* alone, the *offense type* predictor was removed again, but the LSI-R's prediction remained statistically insignificant.

## Discussion

The main aim of this study was to assess whether psychological processing during episodes of criminal behavior can be measured using standardized content analysis of its descriptions in sentences, and whether this approach might be useful in predicting future criminal behavior of convicted offenders. In general, both seem to be the case. While only some of the target variables evidence acceptable inter-rater agreement, 8 out of 10 measures derived from the verdict-based reconstruction of the offense sequence (URD) show associations with at least one of the risk markers and/or criminal reoffending criteria.

When comparing the person-oriented Level of Service Inventory-Revised (LSI-R) and the crime scene-oriented Crime Scene Behavior Risk Measure (CBR), the latter shows, overall, stronger associations with the URD measures.<sup>5</sup> Interestingly, the downregulation of negative affect during criminal behavior as well as perseverance in

<sup>5</sup> It should be noted, however, that the CBR data include only sex offenders. Thus, different samples (and sample sizes) are being compared here.

the face of implementation difficulties was associated with a reduced *criminal history* score (LSI-R subscale 1), which was the only static risk factor included in this study. Apart from these findings, the overall pattern of associations with the LSI-R subscales (Table 2) gives the impression that the more psychologically meaningful an LSI-R risk marker is, the more we find associations with URD measures. For the CBR (Table 3), significant correlations were found for item 1 (*multiple juvenile offenders*) and item 5 (*male victim*).

Not surprisingly, which URD variables predict reoffending depends on the specific criterion: Perseverance negatively predicts both sexual and general reoffending, while negatively motivated impulsivity predicts violent reoffense. These two are the most promising URD variables for prognostic purposes, not only because of their associations with official reoffenses, but also because they appear to be measurable with acceptable inter-rater agreement. In addition, they appear to be incrementally valid with respect to established risk instruments. First, in predicting violent reoffense, negatively motivated impulsivity not only showed incremental predictive validity with respect to the LSI-R, which is a valid instrument for predicting violent reoffending (Barnoski and Aos 2003; Dahle 2006), but it fully mediated its predictive contribution. This is exciting because mediation can indicate causation (e.g., Jung 2021), and reoffense prediction research desperately needs psychologically meaningful mediators (Walters 2017). Indeed, from a theoretical perspective, the most plausible (though not the only viable) explanation for the predictive relationship seems to be that negatively motivated impulsivity may be somewhat intrapersonally stable and, on the other hand, facilitate violent offending.

As for perseverance, which might be interpreted as the strength of determination to carry out an intention illegally, it may be surprising that it reduces rather than increases the risk of (sexual/general) reoffending. However, nonperseverance would also be coded when a lawful intention is hindered by implementation difficulties and the resulting intention-deviant behavior would violate a law. It remains to be investigated which action variant is associated with the risk of sexual and general reoffending. When predicting sexual reoffending, the reference risk assessment tool, the CBR (Dahle et al. 2014), lost its predictive validity to perseverance. This full mediation may be less surprising than the one discussed previously, as the CBR, like the URD, is a measure of the offense itself. However, given that perseverance is only correlated with one CBR item, namely item 5 (*male victim*), the finding might actually further our understanding of the psychology behind crime scene behavior. It may give us an idea as to why such a seemingly irrelevant aspect as the victim's gender predicts future sexual

offending. Further in-depth item-level analyses are needed to clarify the relationships.

Overall, the findings suggest that reoffending with sexual and violent crime may be based on different individual peculiarities in psychological processing. It is not yet clear whether the predictive value found for some of the URD variables results from their potential intraindividual stability from one offense to the next. In any case, the associations seem to reflect more than mere specialization, as they remained unchanged when the index offense type was controlled for. In the future, a more nuanced view of the criterion *offense type* may enhance our understanding of the psychological mechanisms underlying reoffending—although psychologically meaningful categorization of offense types is itself a Herculean task (Ward et al. 2021).

As a measure of the psychological processing as described in criminal judgments, the URD demonstrates the promise of this approach, but also the need for methodological refinement. Inter-coder agreement is poor for most of the variables reported, and even for those which were coded with acceptable agreement, there is room for improvement. Moreover, certain limitations are inherent to the URD methodology. First, the detail of the descriptions in the sentences limits the precision of the URD codings. Verdicts containing less than one page of information about the criminal episode were categorically excluded; these were typically less serious offenses that resulted in a prison sentence due to prior convictions, especially probation. Even among the longer descriptions, the level of detail varies greatly between sentences, and its effect on psychometric parameters remains to be investigated. Second, it is important to remember that the URD, like any content analysis (see Kessler et al. 2023), does not analyze the event described, but the author's reconstruction of it. Implicit theories and attitudes held by judges are known to influence judgments of guilt and punishment (Harper and Bartels 2018; Korkmaz et al. 2024), and are therefore likely to also be reflected in the narrative reconstruction of the offense itself. Third, the episode analyzed is a sample of the individual's natural behavior, selected according to judicial criteria. While this sampling method is advantageous for external validity, its nonexperimental nature poses a significant problem: Not every verdict contains enough information to calculate all the variables of interest. The variables that allow for the most nuanced analysis of successive processing are those that indicate a conditional probability (i.e., the NA, PA, H, and EP variables). These can be considered the most psychologically meaningful of the URD measures. They allow for a process-oriented analysis that systematically takes into account the sequentiality of the action episode, which is a clear advantage over more objective crime scene assessments (e.g., Biedermann 2013; Dahle et al. 2014). However, it is these variables that cannot be derived from every judg-

ment. For instance, *downregulation NA* cannot be assigned a value unless at least one instance of activation via negative affect (NA) is observed.

This systematic production of missing values for some critical variables results in varying sample sizes across analyses. Additionally, for all analyses involving the CBR, only offenders whose “initial” offense (i.e., the offense that led to incarceration) was a sexual offense could be included. Not only do these reductions in sample size affect statistical power, but because of the nonrandom nature of the exclusion, they could introduce bias and make it difficult to interpret the results. To at least partially address this issue, the “unconditional” variants of all variables (i.e., the A variants) were analyzed and reported as well.

Finally, despite the approach being rooted in a criminal-psychological reading of an action-regulation theory (Kuhl 2001; Kuhl et al. 2021), this study tested the predictive value of the URD for reoffending in a purely exploratory manner. Indeed, it should be noted that from a theoretical perspective, strong, uniform main effects of isolated psychological processes on reoffending are not necessarily to be expected (e.g., Anderson and Bushman 2002; Wikström and Treiber 2015; Schmidt and Ward 2021). Instead, the self and affect regulation theory of adherence to social norms (Fig. 1) posits interactions between affective processing and various dispositional and situational norm-enforcing instances (Hamatschek et al. 2022). These complex relationships remain to be explored.

The results reported here indicate that the descriptions of criminal behavior in (German) judgments contain substantial information about the subject’s psychological processing beyond what is currently used. This information could contribute to risk assessment and, potentially, to case management and treatment planning. For instance, it could be utilized in defining the treatment needs and individual responsiveness according to the risk-needs-responsivity model (Bonta and Andrews 2017)—especially since a practical advantage of the method is that it is relatively independent of the subject’s cooperativeness. The URD is currently undergoing revision based on the results of this analysis and the feedback from coders on specific deficits in clarity and instructive gaps. In contrast to the summative measurement of individual traits or contextual conditions, the process-oriented method allows for a dissociation between consecutive, partly conditional processes, thereby providing insights into real-life psychological functioning. The approach may have the potential to enhance nomothetic risk assessment while providing an additional feedback loop for the idiographic prognosis, thus connecting the two. This may contribute to a clearer understanding of the complex interactions leading to criminal (re)offending or norm adherence.

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## Appendix

**Table 5** Illustration of the URD coding procedure

| Verdict text (fictitious)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Code | Calculation of URD variables |            |                      |           |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------|------------|----------------------|-----------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      | SARTAN phases:               | Regulation | Intention generation | Enactment | Result |
| ... After this fight with his father F., the defendant went to his room where he thought about his situation. Finally, he decided that his whole family would be better off without F. and that he was going to kill him. However, he had no concrete plan of realizing this intention right now. He went to the kitchen to find something to eat, when he found a CO <sub>2</sub> cartridge for the soda maker in a shopping bag. He took it with him into the living room where F. was sitting in the coach and hit him with it from the back, targeting his head. However, F. had seen his reflection in the TV and partly dodged the attack; the cartridge had only hit him on the shoulder. He yelled and tried to grab the cartridge, calling the defendant a XX and an XX. For a moment, the defendant was shocked that F. was still alive, yet he did not back off but held tight to the cartridge. The insults, however, enraged him again and he stabbed F. in the belly with the cartridge. ... | N    | Activation                   | B          | ANV/NK               | SRKAP     | ENK    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | P    |                              | KSV+       | K/NK                 | SRKAP     | ENK    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | N    |                              | W/SV+      | K/NV                 | SRAP/FT   | ENV    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | N    |                              |            | IVSv                 | IVSKAP    | ENV    |

Abbreviations in code, from first to last phase: *N* negative activation affect, *P* positive activation affect, *B* successful downregulation of a negative affect, *KSV+* no impulsive reaction, positive affect, *W/SV+* hostile emotions remain after potential regulation of negative affect, impulsive reaction, *A* general intention generated, *K* (only) concrete intention generated, *NV* intention contains norm-violating elements, *NK* intention contains no norm-violating elements, *IVSv* impulsive reaction contains norm-violating elements, *SRKAP* self-regulated action without intention implementation difficulties, *SRAP/FT* an intention implementation problem occurs during self-regulated action and is met with perseverance in the concrete intention, *IVSKAP* impulse reaction without difficulties in enactment, *ENK* result is norm-conform, *ENV* result is norm-violating. The example does not contain all possible URD paths

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## Declarations

**Conflict of interest** M.J. Hamatschek, M.S. Richter and K.-P. Dahle declare that they have no competing interests. One of the coauthors, K.-P. Dahle, is a member of the FPPK's editorial board.

**Ethical standards** All studies whose data were used were approved by the German Psychological Association's (DGPs) research ethics board with respect to the professional ethical guidelines formulated by DGPs and BDP (German Association of Psychologists). The research procedure was reviewed and authorized by the Berlin Ministry of Justice. Due to the retrospective study design, it was not possible to obtain the subjects' consent on the assessment of their records or on their inclusion in the publication. Therefore, for each study to be authorized, a comprehensive data security policy was developed to which all participating researchers pledged adherence in written form. Complete pseudonymization was guaranteed while ensuring reassignability of reconviction data collected at a later time. Conformity with the then-applicable German and European data privacy law was attested by the Berlin Commissioner for Data Protection and Freedom of Information. As for the complementary re-assessment of some file material (verdicts) that was necessary for this study, sensible information was blackened prior to coding. Moreover, coders signed a confidentiality agreement in which they were referred to the applicable professional discretion regulations (specifically, § 203 StGB and § 10 NDSGAQ3).

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